Yonatan Bisk

Leonardo Associate Professor of Computer Science

GHC 6703
✉ *ybisk@cs.cmu.edu*
he/him

Positions

2026 – **Leonardo Associate Professor** *Carnegie Mellon University*
Language Technologies Institute
Robotics Institute (courtesy)
Electrical and Computer Engineering (courtesy)

2026 – **Research Advisor** *Origami Robotics*

Previous

2022 – **Visiting Professor** *Meta Fundamental AI Research (FAIR)* w/ Dhruv Batra & Roozbeh Mottaghi

2021 – **Consulting Professor** *Microsoft Research*

2020 – **Assistant Professor** *Carnegie Mellon University*
Language Technologies Institute

2019 – **Visiting Researcher** *Microsoft Research* w/ Jianfeng Gao

2017 – **Postdoctoral Scholar**
University of Washington & Allen Institute for Artificial Intelligence w/ Yejin Choi

2015 – **Postdoctoral Scholar**
Information Sciences Institute, University of Southern California w/ Daniel Marcu

Education

2015 **PhD in Computer Science** *The University of Illinois at Urbana-Champaign*
Thesis: **Unsupervised Grammar Induction with Combinatory Categorial Grammars**
Advisor: Julia Hockenmaier
Committee: Jason Eisner, Dan Roth, ChengXiang Zhai

2009 **BS in Computer Science** *The University of Texas at Austin* Turing Scholar
Thesis: **The Necessity of Separating Control and Logic When Grounding Language using Neuroevolution**
Advisor: Risto Miikkulainen Second Reader: Raymond Mooney

Postdocs

TalkingToRobots.com

Post	**Maithili Patel**	she/her	*Proactive Robotics*		2026 –

PhD Students

PhD	**Abitha Thankaraj**	she/her	*Language Modeling*	*(co-Zico Kolter)*	CSD 2024 –
PhD	**Rosa Vitiello**	she/her	*Pedagogy*		LTI 2024 –
PhD	**Quanting Xie**	he/him	*Dexterity*		RI 2024 –
PhD	**Leena Mathur**	she/her	*Social Intelligence*	*(co-LP Morency)*	LTI 2024 –
PhD	**Haochen Zhang**	she/her	*Robust Robot Learning*		RI 2025 –
PhD	**Annabella Macaluso**	she/her	*Robot Generalization*		RI 2025 –
PhD	**Wenjie Li**	she/her	*Machine Cognition*		NI 2024 –
PhD	**Jacob Thompson**	he/him	*Spatial Reasoning*		LTI 2026 –
PhD	**Ani Kimmel**	she/her	*3D Manufacturing*	*(co-Gary Fedder)*	ECE 2025 –

Masters

MSR	**Soumya Teotia**	she/her	*Robot ToM*	2025 –

Alumni

PhD

LTI	**Yingshan Chang**				2022-2026
	Generalization Beyond Observed Support				
LTI	**Jared Fernandez** *(co-Emma Strubell)*				2020-2026
	Towards Efficient & Sustainable AI: Workload Characterizations and Algorithmic Mitigations				
RI	**Vidhi Jain**				2021-2025
	Lowering Barriers in Human-Robot Communication				
MLD	**So Yeon Min** *(co-Rus Salakhutdinov)*				2021-2025
	Situated Instruction Following				
LTI	**Hao Zhu** *(co-Graham Neubig)*				2019-2024
	Artificial Social Intelligence: Challenges, Environments, and Mechanisms				

MS

MLT	**Yingshan Chang**	she/her	*Multimodal Reasoning*	LTI	2020 – 2022	
MS	**Quanting Xie**	he/him	*LLMs for Robotics*	MechE	2023	
MSR	**Jacob Thompson**	he/him	*Spatial Reasoning*	RI	2024 – 2026	
MSR	**Yash Jangir**	he/him	*Robot Evaluation*	RI	2024 – 2026	
5th	**Claire Jin**	she/her	*Mapping*		2026	
5th	**Sheryl Mathew**	she/her	*Collaboration*	SCS	2024	
MLT	**Jimin Sun**	she/her	*Embodied Dialogue*	LTI	2022 – 2023	
MLT	**Josh Zhanson**	he/him	*Embodied Visual Reps*	LTI	2020 – 2022	
5th	**Xiaoyu (Erin) Zhang**	she/her	*Collaborative Robots* *(co-Oliver Kroemer)*	5th	2021 – 2022	

Interns and Undergraduates

Intern	**Mimo Shirasaka**		*Embodied Perspective*		2026
Intern	**Minyoung Hwang**	she/her	*Preference Learning*		2024
Ugrad	**Su Li**	she/her	*Robot Learning*		2024
Ugrad	**Maria Yampolsky**	she/her	*Audio*		2024
Ugrad	**Xiang Fan**	he/him	*Transformer Efficiency*	*(co-Emma Strubell)*	2022
Intern	**Andy Liu**	he/him	*Theory-of-Mind*	*(co-Graham Neubig)*	2022
Intern	**Deogratias Amani**	he/him	*Grounding*		2022
Ugrad	**Feiyu (Gavin) Zhu**	he/him	*Video Summarization*	*(co-Chris Paxton)*	2021
Intern	**Blessing Bassey**	she/her	*TextVQA*		2021
Intern	**Melanie Sclar**	she/her	*Symmetric Theory-of-Mind*	*(co-Graham Neubig)*	2021

Capstones

MCDS	**VR Robot Teleop**			2024
	Sophia Huang	Jensen (Yi-Che) Huang	Tianyi Cheng	
MSAII	**Seed emergence**			2024
	Harini Subramanyan	Juanyun Mai	Ameya Morbale	
	Karthik Natarajan	Noel D Souza	Bohao Zhang	
MRSD	**Assistive Robotics for Eldercare** (*co-Z Erickson)			2023
	Abhinav Gupta he/him	Atharva Pusalkar he/him		
	Praveen Venkatesh he/him	Shaolin Kataria he/him		
	Shivam Tripathy he/him			
MRSD	**Modular Teleop with Autonomy** (*co-Z Erickson)			2022
	Jashkumar Diyora he/him	Shruti Gangopadhyay she/her		
	Jigar Patel he/him	Prakhar Pradeep he/him		
	Shivani Sivakumar she/her			
MCDS	**Embodied Navigation and Interaction**			2022
	Adhokshaja Madhwaraj he/him	Kushagra Mahajan he/him		
	Sai Vishwas Padigi he/him	Vineeth Reddy he/him		
	Malaika Vijay she/her	Zheng (Jessica) Zhong she/her		

MCDS **Embodied Dialogue and ToM** 2022
Xinyue Chen she/her, Nikhil Gupta he/him, Benny Jiang he/him, Shubham Phal he/him, Prasoon Varshney he/him

MCDS **Novel Concept Learning** 2022
Bharani Ujjaini Kempaiah she/her, Linyi Li she/her

MSAII **Visual Patent Search** 2022
Suchong Dong she/her, Zhiyuan Guo he/him, Mansi Goyal she/her, Marta Mendez Simon she/her, Sreenidhi Sundaram she/her, Leon Van Den Berghe he/him

Journal Publications

1 **Bridging Language and Action: A Survey of Language-Conditioned Robot Manipulation** 2026
Xiangtong Yao, Hongkuan Zhou, Oier Mees, Yuan Meng, Ted Xiao, Yonatan Bisk, Jean Oh, Edward Johns, Mohit Shridhar, Dhruv Shah, Jesse Thomason, Kai Huang, Joyce Chai, Zhenshan Bing, Alois Knoll"
International Journal of Robotics Research

2 **Looking beyond the next token** 2026
Abitha Thankaraj, Yiding Jiang, J. Zico Kolter, Yonatan Bisk
Transactions on Machine Learning Research

3 **Affordance RAG: Hierarchical Multimodal Retrieval with Affordance-Aware Embodied Memory for Mobile Manipulation** 2025
Ryosuke Korekata, Quanting Xie, Yonatan Bisk, Komei Sugiura
IEEE Robotics and Automation Letters

4 **Efficient Hardware Scaling and Diminishing Returns in Large-Scale Training of Language Models** 2025
Jared Fernandez, Luca Wehrstedt, Leonid Shamis, Mostafa Elhoushi, Kalyan Saladi, Yonatan Bisk, Emma Strubell, Jacob Kahn
Transactions on Machine Learning Research

5 **MotIF: Motion Instruction Fine-tuning** 2025
Minyoung Hwang, Joey Hejna, Dorsa Sadigh, Yonatan Bisk
IEEE Robotics and Automation Letters

6 **Training Vision-Language Transformers from Captions** 2023
Liangke Gui, Yingshan Chang, Qiuyuan Huang, Subhojit Som, Alexander Hauptmann, Jianfeng Gao, Yonatan Bisk
Transactions on Machine Learning Research

7 **The Return of Lexical Dependencies: Neural Lexicalized PCFGs** 2020
Hao Zhu, Yonatan Bisk, Graham Neubig
Transactions of the Association for Computational Linguistics

8 **An HDP Model for Inducing Combinatory Categorial Grammars** 2013
Yonatan Bisk, Julia Hockenmaier
Transactions of the Association for Computational Linguistics

Conference Publications

9 **Position: Anthropomorphic Language in AI Discourse Can Be Constructive** 2026
Yingshan Chang, Yonatan Bisk
NeurIPS 2026 Position Paper Track

10 **Data Augmentation for Improving Contextual Faithfulness Under Knowledge Conflict** 2026
Mohith Rajesh, Raj Maheshwari, Vashisth Tiwari, Yonatan Bisk, Emma Strubell, Jeremiah Milbauer, Jared Fernandez
Findings of The 5th Asia-Pacific Chapter of the Association for Computational Linguistics & the 15th International Joint Conference on Natural Language Processing

11 **Contextual Observer Grounding: Evaluating Situated Spatial Reasoning in Vision-Language Models** 2026
Mimo Shirasaka, Haochen Zhang, Yonatan Bisk
Findings of The 2026 Conference on Empirical Methods in Natural Language Processing

12 **Amplified Does Not Mean Predictive: Reasoning Behaviors in Thinking Models** 2026
Jean de Dieu Nyandwi, Leena Mathur, Yonatan Bisk, Graham Neubig
Third Conference on Language Modeling

13 **IndoorR2X: Indoor Robot-to-Everything Coordination with LLM-Driven Planning** 2026
Fan Yang, Soumya Teotia, Shaunak A. Mehta, Prajit KrisshnaKumar, Quanting Xie, Jun Liu, Yueqi Song, Wenkai Li, Atsunori Moteki, Kanji Uchino, Yonatan Bisk
IEEE/RSJ International Conference on Intelligent Robots and Systems

14 **Position: Evaluation of ML Resource Utilization Requires Model Life Cycle Assessment** 2026
Jared Fernandez, Clara Na, Yonatan Bisk, Constantine Samaras, Emma Strubell
Forty-third International Conference on Machine Learning Position Paper Track

15 **MM-SeR: Multimodal Self-Refinement for Lightweight Image Captioning** 2026
Junha Song, Yongsik Jo, So Yeon Min, Quanting Xie, Taehwan Kim, Yonatan Bisk, Jaegul Choo
Computer Vision and Pattern Recognition Conference

16 **RobotArena ∞: Scalable Robot Benchmarking via Real-to-Sim Translation** 2026
Yash Jangir, Yidi Zhang, Kashu Yamazaki, Chenyu Zhang, Kuan-Hsun Tu, Tsung-Wei Ke, Lei Ke, Yonatan Bisk, Katerina Fragkiadaki
The Fourteenth International Conference on Learning Representations

17 **Unsupervised Discovery of Long-Term Spatiotemporal Periodic Workflows in Human Activities** 2026
Fan Yang, Quanting Xie, Atsunori Moteki, Shoichi Masui, Shan Jiang, Kanji Uchino, Yonatan Bisk, Graham Neubig
The Winter Conference on Applications of Computer Vision

18 **RAVEN: Resilient Aerial Navigation via Open-Set Semantic Memory and Behavior Adaptation** 2026
Seungchan Kim, Omar Alama, Dmytro Kurdydyk, John Keller, Nikhil Keetha, Wenshan Wang, Yonatan Bisk, Sebastian Scherer
2026 IEEE International Conference on Robotics and Automation (ICRA)

19 **FieldWorkArena: Agentic AI Benchmark for Real Field Work Tasks** 2026
Atsunori Moteki, Shoichi Masui, Fan Yang, Yueqi Song, Yonatan Bisk, Graham Neubig, Ikuo Kusajima, Yasuto Watanabe, Hiroyuki Ishida, Jun Takahashi, Shan Jiang
International Conference on Pattern Recognition

20 **REM: Evaluating LLM Embodied Spatial Reasoning through Multi-Frame Trajectories** 2025
Jacob Thompson, Emiliano Garcia-Lopez, Yonatan Bisk
Second Conference on Language Modeling

21 **MolErr2Fix: Benchmarking LLM Trustworthiness in Chemistry via Modular Error Detection, Localization, Explanation, and Correction** 2025
Yuyang Wu, Jinhui Ye, Shuhao Zhang, Lu Dai, Yonatan Bisk, Olexandr Isayev
Conference on Empirical Methods in Natural Language Processing

22 **Position: You Can't Manufacture a NeRF** 2025
Ani Kimmel, Mueed Ur Rehman, Yonatan Bisk, Gary K. Fedder
Forty-second International Conference on Machine Learning Position Paper Track

23 **Casper: Inferring Diverse Intents for Assistive Teleoperation with Vision Language Models** 2025
Huihan Liu, Rutav Shah, Shuijing Liu, Jack Pittenger, Mingyo Seo, Yuchen Cui, Yonatan Bisk, Roberto Martín-Martín, Yuke Zhu
Conference on Robot Learning

24 **Energy Considerations of Large Language Model Inference and Efficiency Optimizations** 2025
Jared Fernandez, Clara Na, Vashisth Tiwari, Yonatan Bisk, Sasha Luccioni, Emma Strubell
The 63rd Annual Meeting of the Association for Computational Linguistics

25 **Re-thinking Temporal Search for Long-Form Video Understanding** 2025
Jinhui Ye, Zihan Wang, Haosen Sun, Keshigeyan Chandrasegaran, Zane Durante, Cristobal Eyzaguirre, Yonatan Bisk, Juan Carlos Niebles, Ehsan Adeli, Li Fei-Fei, Jiajun Wu, Manling Li
The IEEE/CVF Conference on Computer Vision and Pattern Recognition

26 **Explore Theory-of-Mind: Program-Guided Adversarial Data Generation for Theory of Mind Reasoning** 2025
Melanie Sclar, Jane Yu, Maryam Fazel-Zarandi, Yulia Tsvetkov, Yonatan Bisk, Yejin Choi, Asli Celikyilmaz
The Thirteenth International Conference on Learning Representations

27 **Language Models Need Inductive Biases to Count Inductively** 2025
Yingshan Chang, Yonatan Bisk
The Thirteenth International Conference on Learning Representations

28 **Direct Preference Optimization of Video Large Multimodal Models from Language Model Reward** 2025
Ruohong Zhang, Liangke Gui, Zhiqing Sun, Yihao Feng, Keyang Xu, Yuanhan Zhang, Di Fu, Chunyuan Li, Alexander Hauptmann, Yonatan Bisk, Yiming Yang
Annual Conference of the Nations of the Americas Chapter of the ACL

29 **ANAVI: Audio Noise Awareness using Visuals of Indoor environments for NAVIgation** 2024
Vidhi Jain, Rishi Veerapaneni, Yonatan Bisk
Conference on Robot Learning

30 **Situated Instruction Following** 2024
So Yeon Min, Xavi Puig, Devendra Singh Chaplot, Tsung-Yen Yang, Akshara Rai, Priyam Parashar, Ruslan Salakhutdinov, Yonatan Bisk, Roozbeh Mottaghi
European Conference on Computer Vision

31 **Gradient Localization Improves Lifelong Pretraining of Language Models** 2024
Jared Fernandez, Yonatan Bisk, Emma Strubell
Findings of the Conference on Empirical Methods in Natural Language Processing

32 **How to Train Your Fact Verifier: Knowledge Transfer with Multimodal Open Models** 2024
Jaeyoung Lee, Ximing Lu, Jack Hessel, Faeze Brahman, Youngjae Yu, Yonatan Bisk, Yejin Choi, Saadia Gabriel
Findings of the Conference on Empirical Methods in Natural Language Processing

33 **Tools Fail: Detecting Silent Errors in Faulty Tools** 2024
Jimin Sun, So Yeon Min, Yingshan Chang, Yonatan Bisk
Conference on Empirical Methods in Natural Language Processing

34 **DiffusionPID: Interpreting Diffusion via Partial Information Decomposition** 2024
Shaurya Dewan, Rushikesh Zawar, Prakanshul Saxena, Yingshan Chang, Andrew Luo, Yonatan Bisk
Thirty-Eighth Annual Conference on Neural Information Processing Systems

35 **VISREAS: Complex Visual Reasoning with Unanswerable Questions** 2024
Syeda Nahida Akter, Sangwu Lee, Yingshan Chang, Yonatan Bisk, Eric Nyberg
Findings of the 62nd Annual Meeting of the Association for Computational Linguistics

36 **AgentKit: Flow Engineering with Graphs, not Coding** 2024
Yue Wu, Yewen Fan, So Yeon Min, Shrimai Prabhumoye, Stephen McAleer, Yonatan Bisk, Ruslan Salakhutdinov, Yuanzhi Li, Tom Mitchell
Conference on Language Modeling

37 **Skews in the Phenomenon Space Hinder Generalization in Text-to-Image Generation** 2024
Yingshan Chang, Yasi Zhang, Zhiyuan Fang, Yingnian Wu, Yonatan Bisk, Feng Gao
European Conference on Computer Vision

38 **Vid2Robot: End-to-end Video-conditioned Policy Learning with Cross-Attention Transformers** 2024
Vidhi Jain, Maria Attarian, Nikhil J Joshi, Ayzaan Wahid, Danny Driess, Quan Vuong, Pannag R Sanketi, Pierre Sermanet, Stefan Welker, Christine Chan, Igor Gilitschenski, Yonatan Bisk, Debidatta Dwibedi
Robotics: Science and Systems

39 **SOTOPIA-π: Interactive Learning of Socially Intelligent Language Agents** 2024
Ruiyi Wang, Haofei Yu, Wenxin Zhang, Zhengyang Qi, Maarten Sap, Graham Neubig, Yonatan Bisk, Hao Zhu
The 62nd Annual Meeting of the Association for Computational Linguistics

40 **OpenEQA: Embodied Question Answering in the Era of Foundation Models** 2024
Arjun Majumdar, Anurag Ajay, Xiaohan Zhang, Pranav Putta, Sriram Yenamandra, Mikael Henaff, Sneha Silwal, Paul Mcvay, Oleksandr Maksymets, Sergio Arnaud, Karmesh Yadav, Qiyang Li, Ben Newman, Mohit Sharma, Vincent Berges, Shiqi Zhang, Pulkit Agrawal, Yonatan Bisk, Dhruv Batra, Mrinal Kalakrishnan, Franziska Meier, Chris Paxton, Sasha Sax, Aravind Rajeswaran
Conference on Computer Vision and Pattern Recognition

41 **SOTOPIA: Interactive Evaluation for Social Intelligence in Language Agents** 2024
Xuhui Zhou, Hao Zhu, Leena Mathur, Ruohong Zhang, Haofei Yu, Zhengyang Qi, Louis-Philippe Morency, Yonatan Bisk, Daniel Fried, Graham Neubig, Maarten Sap
The Twelfth International Conference on Learning Representations (Spotlight)

42 **Open X-Embodiment: Robotic Learning Datasets and RT-X Models: Open X-Embodiment Collaboration** 2024
about 300 people... including me and Vidhi
IEEE International Conference on Robotics and Automation (ICRA)

43 **MOSAIC: Learning Unified Multi-Sensory Object Property Representations for Robot Perception** 2024
Gyan Tatiya, Jonathan Francis, Ho-Hsiang Wu, Yonatan Bisk, Jivko Sinapov
IEEE International Conference on Robotics and Automation

44 **WebArena: A Realistic Web Environment for Building Autonomous Agents** 2024
Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, Graham Neubig
The Twelfth International Conference on Learning Representations

45 **Spatial-Language Attention Policies for Efficient Robot Learning** 2023
Priyam Parashar, Vidhi Jain, Xiaohan Zhang, Jay Vakil, Sam Powers, Yonatan Bisk, Chris Paxton
Conference on Robot Learning

46 **SPAE: Semantic Pyramid AutoEncoder for Multimodal Generation with Frozen LLMs** 2023
Lijun Yu, Yong Cheng, Zhiruo Wang, Vivek Kumar, Wolfgang Macherey, Yanping Huang, David A. Ross, Irfan Essa, Yonatan Bisk, Ming-Hsuan Yang, Kevin Murphy, Alexander Hauptmann, Lu Jiang
Thirty-seventh Conference on Neural Information Processing Systems

47 **HomeRobot: Open-Vocabulary Mobile Manipulation** 2023
Sriram Yenamandra, Arun Ramachandran, Karmesh Yadav, Austin Wang, Mukul Khanna, Theophile Gervet, Tsung-Yen Yang, Vidhi Jain, Alexander William Clegg, John Turner, Zsolt Kira, Manolis Savva, Angel Chang, Devendra Singh Chaplot, Dhruv Batra, Roozbeh Mottaghi, Yonatan Bisk, Chris Paxton
Conference on Robot Learning

48 **SPRING: GPT-4 Out-performs RL Algorithms by Studying Papers and Reasoning** 2023
Yue Wu, Shrimai Prabhumoye, So Yeon Min, Yonatan Bisk, Ruslan Salakhutdinov, Amos Azaria, Tom Mitchell, Yuanzhi Li
Thirty-seventh Conference on Neural Information Processing Systems

49 **Towards Open-World Mobile Manipulation in Homes: Lessons from the Neurips 2023 HomeRobot Open Vocabulary Mobile Manipulation Challenge** 2023
Sriram Yenamandra, Arun Ramachandran, Mukul Khanna, Karmesh Yadav, Jay Vakil, Andrew Melnik, Michael Büttner, Leon Harz, Lyon Brown, Gora Chand Nandi, Arjun PS, Gaurav Kumar Yadav, Rahul Kala, Robert Haschke, Yang Luo, Jinxin Zhu, Yansen Han, Bingyi Lu, Xuan Gu, Qinyuan Liu, Yaping Zhao, Qiting Ye, Chenxiao Dou, Yansong Chua, Volodymyr Kuzma, Vladyslav Humennyy, Ruslan Partsey, Jonathan Francis, Devendra Singh Chaplot, Gunjan Chhablani, Alexander Clegg, Theophile Gervet, Vidhi Jain, Ram Ramrakhya, Andrew Szot, Austin Wang, Tsung-Yen Yang, Aaron Edsinger, Charlie Kemp, Binit Shah, Zsolt Kira, Dhruv Batra, Roozbeh Mottaghi, Yonatan Bisk, Chris Paxton
Thirty-seventh Conference on Neural Information Processing Systems: Competition Track

50 **EXCALIBUR: Encouraging and Evaluating Embodied Exploration** 2023
Hao Zhu, Raghav Kapoor, So Yeon Min, Winson Han, Jiatai Li, Kaiwen Geng, Graham Neubig, Yonatan Bisk, Aniruddha Kembhavi, Luca Weihs
Conference on Computer Vision and Pattern Recognition

51 **The Framework Tax: Disparities Between Inference Efficiency in Research and Deployment** 2023
Jared Fernandez, Jacob Kahn, Clara Na, Yonatan Bisk, Emma Strubell
The 2023 Conference on Empirical Methods in Natural Language Processing (Oral)

52 **Computational Language Acquisition with Theory of Mind** 2023
Andy Liu, Hao Zhu, Emmy Liu, Yonatan Bisk, Graham Neubig
The Eleventh International Conference on Learning Representations

53 **Self-Supervised Object Goal Navigation with In-Situ Finetuning** 2023
So Yeon Min, Yao-Hung Hubert Tsai, Wei Ding, Ali Farhadi, Ruslan Salakhutdinov, Yonatan Bisk, Jian Zhang
IEEE/RSJ International Conference on Intelligent Robots and Systems

54 **Don't Copy the Teacher: Data and Model Challenges in Embodied Dialogue** 2022
So Yeon Min, Hao Zhu, Ruslan Salakhutdinov, Yonatan Bisk
Conference on Empirical Methods in Natural Language Processing

55 **EvEntS ReaLM: Event Reasoning of Entity States via Language Models** 2022
Evangelia Spiliopoulou, Artidoro Pagnoni, Yonatan Bisk, Eduard Hovy
Conference on Empirical Methods in Natural Language Processing

56 **On Advances in Text Generation from Images Beyond Captioning: A Case Study in Self-Rationalization** 2022
Shruti Palaskar, Akshita Bhagia, Yonatan Bisk, Florian Metze, Alan W Black, Ana Marasovic
Findings of the Conference on Empirical Methods in Natural Language Processing

57 **Transformers are Adaptable Task Planners** 2022
Vidhi Jain, Yixin Lin, Eric Undersander, Yonatan Bisk, Akshara Rai
Conference on Robot Learning

58 **Symmetric Machine Theory of Mind** 2022
Melanie Sclar, Graham Neubig, Yonatan Bisk
39th International Conference on Machine Learning

59 **A Framework for Learning to Request Rich and Contextually Useful Information from Humans** 2022
Khanh Nguyen, Yonatan Bisk, Hal Daume III
39th International Conference on Machine Learning,

60 **Simulated Language Learning from Communicative Goals and Linguistic Input** 2022
Hao Zhu, Yonatan Bisk, Graham Neubig
Annual Meeting of the Cognitive Science Society (Oral)

61 **HEAR 2021: Holistic Evaluation of Audio Representations** 2022
Joseph Turian, Jordie Shier, Humair Raj Khan, Bhiksha Raj, Björn W. Schuller, Christian J. Steinmetz, Colin Malloy, George Tzanetakis, Gissel Velarde, Kirk McNally, Max Henry, Nicolas Pinto, Camille Noufi, Christian Clough, Dorien Herremans, Eduardo Fonseca, Jesse Engel, Justin Salamon, Philippe Esling, Pranay Manocha, Shinji Watanabe, Zeyu Jin, Yonatan Bisk
Proceedings of Machine Learning Research (PMLR): NeurIPS 2021 Competition Track

62 **KAT: A Knowledge Augmented Transformer for Vision-and-Language** 2022
Liangke Gui, Borui Wang, Qiuyuan Huang, Alexander Hauptmann, Yonatan Bisk, Jianfeng Gao
Annual Conference of the North American Chapter of the Association for Computational Linguistics (Oral)

63 **FILM: Following Instructions in Language with Modular Methods** 2022
So Yeon Min, Devendra Singh Chaplot, Pradeep Ravikumar, Yonatan Bisk, Ruslan Salakhutdinov
The Tenth International Conference on Learning Representations

64 **WebQA: Multihop and Multimodal QA** 2022
Yingshan Chang, Mridu Narang, Hisami Suzuki, Guihong Cao, Jianfeng Gao, Yonatan Bisk
Conference on Computer Vision and Pattern Recognition (Oral)

65 **Dependency Induction Through the Lens of Visual Perception** 2021
Ruisi Su, Shruti Rijhwani, Hao Zhu, Junxian He, Xinyu Wang, Yonatan Bisk, Graham Neubig
The SIGNLL Conference on Computational Natural Language Learning

66 **Language Grounding with 3D Objects** 2021
Jesse Thomason, Mohit Shridhar, Yonatan Bisk, Chris Paxton, Luke Zettlemoyer
Conference on Robot Learning

67 **TACo: Token-aware Cascade Contrastive Learning for Video-Text Alignment** 2021
Jianwei Yang, Yonatan Bisk, Jianfeng Gao
International Conference on Computer Vision

68 **Few-shot Language Coordination by Modeling Theory of Mind** 2021
Hao Zhu, Graham Neubig, Yonatan Bisk
The Thirty-eighth International Conference on Machine Learning

69 **Grounding 'Grounding' in NLP** 2021
Khyathi Raghavi Chandu, Yonatan Bisk, Alan W Black
Findings of The 2021 Conference of the Association for Computational Linguistics

70 **An Empirical Study on the Generalization Power of Neural Representations Learned via Visual Guessing Games** 2021
Alessandro Suglia, Yonatan Bisk, Ioannis Konstas, Antonio Vergari, Emanuele Bastianelli, Andrea Vanzo, Oliver Lemon
The 2021 Conference of the European Chapter of the Association for Computational Linguistics (Oral)

71 **Knowledge-driven Data Construction for Zero-shot Evaluation in Commonsense Question Answering** 2021
Kaixin Ma, Filip Ilievski, Jonathan Francis, Yonatan Bisk, Eric Nyberg, Alessandro Oltramari
35th AAAI Conference on Artificial Intelligence

72 **ALFWorld: Aligning Text and Embodied Environments for Interactive Learning** 2021
Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Côté, Yonatan Bisk, Adam Trischler, Matthew Hausknecht
International Conference on Learning Representations

73 **Imagining Grounded Conceptual Representations from Perceptual Information in Situated Guessing Games** 2020
Alessandro Suglia, Antonio Vergari, Ioannis Konstas, Yonatan Bisk, Emanuele Bastianelli, Andrea Vanzo, Oliver Lemon
The 28th International Conference on Computational Linguistics (Oral)

74 **Experience Grounds Language** 2020
Yonatan Bisk, Ari Holtzman, Jesse Thomason, Jacob Andreas, Yoshua Bengio, Joyce Chai, Mirella Lapata, Angeliki Lazaridou, Jonathan May, Aleksandr Nisnevich, Nicolas Pinto, Joseph Turian
Conference on Empirical Methods in Natural Language Processing (Oral)

75 **RMM: A Recursive Mental Model for Dialog Navigation** 2020
Homero Roman Roman, Yonatan Bisk, Jesse Thomason, Asli Celikyilmaz, Jianfeng Gao
Conference on Empirical Methods in Natural Language Processing: Findings

76 **ALFRED: A Benchmark for Interpreting Grounded Instructions for Everyday Tasks** 2020
Mohit Shridhar, Jesse Thomason, Daniel Gordon, Yonatan Bisk, Winson Han, Roozbeh Mottaghi, Luke Zettlemoyer, Dieter Fox
The IEEE Conference on Computer Vision and Pattern Recognition (CVPR)

77 **PIQA: Reasoning about Physical Commonsense in Natural Language** 2020
Yonatan Bisk, Rowan Zellers, Ronan Le Bras, Jianfeng Gao, Yejin Choi
Thirty-Fourth AAAI Conference on Artificial Intelligence (Oral)

78 **Robust Navigation with Language Pretraining and Stochastic Sampling** 2019
Xiujun Li, Chunyuan Li, Qiaolin Xia, Yonatan Bisk, Asli Celikyilmaz, Jianfeng Gao, Noah A. Smith, Yejin Choi
Conference on Empirical Methods in Natural Language Processing and 9th International Joint Conference on Natural Language Processing

79 **Defending Against Neural Fake News** 2019
Rowan Zellers, Ari Holtzman, Hannah Rashkin, Yonatan Bisk, Ali Farhadi, Franziska Roesner, Yejin Choi
Thirty-third Conference on Neural Information Processing Systems

80 **Improving Robot Success Detection using Static Object Data** 2019
Rosario Scalise, Jesse Thomason, Yonatan Bisk, Siddhartha Srinivasa
IEEE/RSJ International Conference on Intelligent Robots and Systems

81 **Early Fusion for Goal Directed Robotic Vision** 2019
Aaron Walsman, Yonatan Bisk, Saadia Gabriel, Dipendra Misra, Yoav Artzi, Yejin Choi, Dieter Fox
2019 IEEE/RSJ International Conference on Intelligent Robots and Systems
– ***RoboCup Best Paper Award finalist***

82 **HellaSwag: Can a Machine Really Finish Your Sentence?** 2019
Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, Yejin Choi
Association for Computational Linguistics

83 **From Recognition to Cognition: Visual Commonsense Reasoning** 2019
Rowan Zellers, Yonatan Bisk, Ali Farhadi, Yejin Choi
The IEEE Conference on Computer Vision and Pattern Recognition (Oral)

84 **Tactical Rewind: Self-Correction via Backtracking in Vision-and-Language Navigation** 2019
Liyiming Ke, Xiujun Li, Yonatan Bisk, Ari Holtzman, Zhe Gan, Jingjing Liu, Jianfeng Gao, Yejin Choi, Siddhartha Srinivasa
The IEEE Conference on Computer Vision and Pattern Recognition (Oral)

85 **Prospection: Interpretable Plans From Language By Predicting the Future** 2019
Chris Paxton, Yonatan Bisk, Jesse Thomason, Arunkumar Byravan, Dieter Fox
International Conference on Robotics and Automation (ICRA)

86 **Shifting the Baseline: Single Modality Performance on Visual Navigation & QA** 2019
Jesse Thomason, Daniel Gordon, Yonatan Bisk
Annual Conference of the North American Chapter of the Association for Computational Linguistics

87 **Benchmarking Hierarchical Script Knowledge** 2019
Yonatan Bisk, Jan Buys, Karl Pichotta, Yejin Choi
Annual Conference of the North American Chapter of the Association for Computational Linguistics (Oral)

88 **Character-based Surprisal as a Model of Reading Difficulty in the Presence of Errors** 2019
Michael Hahn, Frank Keller, Yonatan Bisk, Yonatan Belinkov
The 41st Annual Meeting of the Cognitive Science Society (Oral)

89 **SWAG: A Large-Scale Adversarial Dataset for Grounded Commonsense Inference** 2018
Rowan Zellers, Yonatan Bisk, Roy Schwartz, Yejin Choi
Conference on Empirical Methods in Natural Language Processing (Oral)

90 **Synthetic and Natural Noise Both Break Neural Machine Translation** 2018
Yonatan Belinkov, Yonatan Bisk
6th International Conference on Learning Representations (Oral)

91 **Learning Interpretable Spatial Operations in a Rich 3D Blocks World** 2018
Yonatan Bisk, Kevin Shih, Yejin Choi, Daniel Marcu
Thirty-Second Conference on Artificial Intelligence (AAAI-18)

92 **Natural Language Inference from Multiple Premises** 2017
Alice Lai, Yonatan Bisk, Julia Hockenmaier
Eighth International Joint Conference on Natural Language Processing (Oral)

93 **Natural Language Communication with Robots** 2016
Yonatan Bisk, Deniz Yuret, Daniel Marcu
15th Annual Conference of the North American Chapter of the Association for Computational Linguistics (Oral)

94 **Supertagging with LSTMs** 2016
Ashish Vaswani, Yonatan Bisk, Kenji Sagae, Ryan Musa
15th Annual Conference of the North American Chapter of the Association for Computational Linguistics (Short Papers) (Oral)

95 **Evaluating Induced CCG Parsers on Grounded Semantic Parsing** 2016
Yonatan Bisk, Siva Reddy, John Blitzer, Julia Hockenmaier, Mark Steedman
2016 Conference on Empirical Methods in Natural Language Processing

96 **Probing the Linguistic Strengths and Limitations of Unsupervised Grammar Induction** 2015
Yonatan Bisk, Julia Hockenmaier
53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Volume 1: Long Papers) (Oral)

97 **Labeled Grammar Induction with Minimal Supervision** 2015
Yonatan Bisk, Christos Christodoulopoulos, Julia Hockenmaier
53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Volume 2: Short Papers)

98 **Simple Robust Grammar Induction with Combinatory Categorial Grammar** 2012
Yonatan Bisk, Julia Hockenmaier
Twenty-Sixth Conference on Artificial Intelligence (AAAI-12)

99 **Document-Topic Hierarchies from Document Graphs** 2012
Tim Weninger, Yonatan Bisk, Jiawei Han
21st ACM international conference on Information and knowledge management (CIKM 2012)

100 **Normal-form parsing for CCGs with generalized composition and type-raising** 2010
Julia Hockenmaier, Yonatan Bisk
23rd International Conference on Computational Linguistics (Coling 2010)

Workshop Publications

101 **MAEA: Multimodal Attribution for Embodied AI** 2022
Vidhi Jain, Jayant Sravan Tamarapalli, Sahiti Yerramilli, Yonatan Bisk
Progress and Challenges in Building Trustworthy Embodied AI at NeurIPS 2022

102 **Tackling AlfWorld with Action Attention and Common Sense from Language Models** 2022
Yue Wu, So Yeon Min, Yonatan Bisk, Ruslan Salakhutdinov, Shrimai Prabhumoye
Language and Reinforcement Learning Workshop at NeurIPS 2022

103 **Worst of Both Worlds: Biases Compound in Pre-trained Vision-and-Language Models** 2022
Tejas Srinivasan, Yonatan Bisk
4th Workshop on Gender Bias in Natural Language Processing

104 **KB-VLP: Knowledge Based Vision and Language Pretraining** 2021
Kezhen Chen, Qiuyuan Huang, Yonatan Bisk, Daniel McDuff, Jianfeng Gao
ICML21 Workshop on Self-Supervised Learning

105 **A Benchmark for Structured Procedural Knowledge Extraction from Cooking Videos** 2020
Frank F. Xu, Lei Ji, Botian Shi, Junyi Du, Graham Neubig, Yonatan Bisk, Nan Duan
Workshop on NLP Beyond Text

106 **Inducing Grammars with and for Neural Machine Translation** 2018
Ke Tran, Yonatan Bisk
2nd Workshop on Neural Machine Translation (Best Paper)

107 **Bridging HMMs and RNNs through Architectural Transformations** 2018
Jan Buys, Yonatan Bisk, Yejin Choi
32nd Conference on Neural Information Processing Systems (NIPS 2018), IRASL workshop

108 **Towards a Dataset for Human Computer Communication via Grounded Language Acquisition** 2016
Yonatan Bisk, Daniel Marcu, William Wong
AAAI 2016 Workshop on Symbiotic Cognitive Systems

109 **Unsupervised Neural Hidden Markov Models** 2016
Ke Tran, Yonatan Bisk, Ashish Vaswani, Daniel Marcu, Kevin Knight
Workshop on Structured Prediction for NLP (Oral)

110 **Induction of Linguistic Structure with Combinatory Categorial Grammars** 2012
Yonatan Bisk, Julia Hockenmaier
NAACL HLT Workshop on Induction of Linguistic Structure

Preprint Publications

111 **ECHO: Embodied Camera Observations of Human Object Carrying** 2026
Xuefei Sun, Lorin Achey, Kali Hamilton, Alberto Speranzon, Gregory Grebe, Yonatan Bisk, Christoffer Heckman
ArXiv

112 **EmbodiedMidtrain: Bridging the Gap between Vision-Language Models and Vision-Language-Action Models via Mid-training** 2026
Yiyang Du, Zhanqiu Guo, Haochen Zhang, Annabella Macaluso, Xin Ye, Liu Ren, Yonatan Bisk, Chenyan Xiong
ArXiv

113 **Inductive Generalization for Robotic Manipulation** 2026
Annabella Macaluso, Haochen Zhang, Ishaan Masilamony, Yingshan Chang, Yonatan Bisk
ArXiv

114 **Lost in Reconstruction: Aligning Action Representations with Language in Vision-Language-Action Models** 2026
Li Wenjie, Yash Jangir, Ignacy Stepka, Yash Agarwal, Marion Kipsang, Yonatan Bisk
ArXiv

115 **Flatness Preserves Instruction Following in Vision-Language-Action Models** 2026
Haochen Zhang, Yonatan Bisk
ArXiv

116 **ELASTIC: Efficiently Learning to Adaptively Scale Test-Time Compute for Generative Control Policies** 2026
Andrew Zou Li, Gokul Swamy, Yonatan Bisk, Andrea Bajcsy
ArXiv

117 **When Should AI Read the Room? Public Perceptions of Social Intelligence in AI Agents** 2026
Leena Mathur, Jenny T. Liang, Vasudha Varadarajan, Jimin Mun, Xuhui Zhou, Jana Schaich Borg, Yonatan Bisk, Louis-Philippe Morency, Maarten Sap
ArXiv

118 **Model Successor Functions** 2025
Yingshan Chang, Yonatan Bisk
ArXiv

119 **Self-Regulation and Requesting Interventions** 2025
So Yeon Min, Yue Wu, Jimin Sun, Max Kaufmann, Fahim Tajwar, Yonatan Bisk, Ruslan Salakhutdinov
ArXiv

120 **Embodied-RAG: General non-parametric Embodied Memory for Retrieval and Generation** 2024
Quanting Xie, So Yeon Min, Tianyi Zhang, Aarav Bajaj, Ruslan Salakhutdinov, Matthew Johnson-Roberson, Yonatan Bisk
ArXiv

121 **DegustaBot: Zero-Shot Visual Preference Estimation for Personalized Multi-Object Rearrangement** 2024
Benjamin A. Newman, Pranay Gupta, Kris Kitani, Yonatan Bisk, Henny Admoni, Chris Paxton
ArXiv

122 **Dialogue with Robots: Proposals for Broadening Participation and Research in the SLIVAR Community** 2024
Casey Kennington, Malihe Alikhani, Heather Pon-Barry, Katherine Atwell, Yonatan Bisk, Daniel Fried, Felix Gervits, Zhao Han, Mert Inan, Michael Johnston, Raj Korpan, Diane Litman, Matthew Marge, Cynthia Matuszek, Ross Mead, Shiwali Mohan, Raymond Mooney, Natalie Parde, Jivko Sinapov, Angela Stewart, Matthew Stone, Stefanie Tellex, Tom Williams
ArXiv

123 **Toward General-Purpose Robots via Foundation Models: A Survey and Meta-Analysis** 2023
Yafei Hu, Quanting Xie, Vidhi Jain, Jonathan Francis, Jay Patrikar, Nikhil Keetha, Seungchan Kim, Yaqi Xie, Tianyi Zhang, Shibo Zhao, Yu Quan Chong, Chen Wang, Katia Sycara, Matthew Johnson-Roberson, Dhruv Batra, Xiaolong Wang, Sebastian Scherer, Zsolt Kira, Fei Xia, Yonatan Bisk
ArXiv

124 **Reasoning about the Unseen for Efficient Outdoor Object Navigation** 2023
Quanting Xie, Tianyi Zhang, Kedi Xu, Matthew Johnson-Roberson, Yonatan Bisk
ArXiv

125 **Plan, Eliminate, and Track – Language Models are Good Teachers for Embodied Agents** 2023
Yue Wu, So Yeon Min, Yonatan Bisk, Ruslan Salakhutdinov, Amos Azaria, Yuanzhi Li, Tom Mitchell, Shrimai Prabhumoye
ArXiv

126 **Retrospectives on the Embodied AI Workshop** 2022
Matt Deitke, Dhruv Batra, Yonatan Bisk, Tommaso Campari, Angel X. Chang, Devendra Singh Chaplot, Changan Chen, Claudia Pérez D'Arpino, Kiana Ehsani, Ali Farhadi, Li Fei-Fei, Anthony Francis, Chuang Gan, Kristen Grauman, David Hall, Winson Han, Unnat Jain, Aniruddha Kembhavi, Jacob Krantz, Stefan Lee, Chengshu Li, Sagnik Majumder, Oleksandr Maksymets, Roberto Martín-Martín, Roozbeh Mottaghi, Sonia Raychaudhuri, Mike Roberts, Silvio Savarese, Manolis Savva, Mohit Shridhar, Niko Sünderhauf, Andrew Szot, Ben Talbot, Joshua B. Tenenbaum, Jesse Thomason, Alexander Toshev, Joanne Truong, Luca Weihs, Jiajun Wu
ArXiv

127 **Multi-View Learning for Vision-and-Language Navigation** 2020
Qiaolin Xia, Xiujun Li, Chunyuan Li, Yonatan Bisk, Zhifang Sui, Jianfeng Gao, Yejin Choi, Noah A. Smith
ArXiv

128 **FIND: Identifying Functionally and Structurally Important Features in Protein Sequences with Deep Neural Networks** 2019
Ranjani Murali, James Hemp, Victoria Orphan, Yonatan Bisk
bioRxiv

129 **Balancing Shared Autonomy with Human-Robot Communication** 2018
Rosario Scalise, Yonatan Bisk, Maxwell Forbes, Daqing Yi, Yejin Choi, Siddhartha Srinivasa
ArXiv

130 **CHALET: Cornell House Agent Learning Environment** 2018
Claudia Yan, Dipendra Misra, Andrew Bennett, Aaron Walsman, Yonatan Bisk, Yoav Artzi
ArXiv Preprint 1801.07357

Thesis Committees

–	**PhD** *Albert Yu*	Raymond Mooney and Roberto Martín-Martín, UT Austin
–	**PhD** *Mark Lee*	Oliver Kroemer, CMU
–	**PhD** *Seungchan Kim*	Sebastian Scherer, CMU
2026	**PhD** *Huihan Liu*	Yuke Zhu, UT Austin
2026	**PhD** *Brendon Boldt*	David Mortensen, CMU
2025	**MSR** *Haochen Zhang*	Wenshan Wang and Ji Zhang, CMU
2025	**PhD** *Liangke Gui*	Alex Hauptmann, CMU
2025	**PhD** *Gabe Sarch*	Katerina Fragkiadaki and Mike Tarr, CMU
2025	**PhD** *Nikolaos Gkanatsios*	Katerina Fragkiadaki, CMU
2024	**PhD** *Xiaoyu Zhu*	Alex Hauptmann, CMU
2024	**PhD** *Yijun Qian*	Alex Hauptmann, CMU
2024	**PhD** *Jielin Qiu*	Lei Li & Christos Faloutsos, CMU
2024	**PhD** *Nikita Srivatsan*	Taylor Berg-Kirkpatrick, CMU
2024	**PhD** *Lijun Yu*	Alex Hauptmann, CMU
2023	**PhD** *Kaixin Ma*	Eric Nyberg, CMU
2023	**PhD** *Nicholas Botzer*	Tim Weninger, Notre Dame
2023	**PhD** *Naoki Otani*	Ed Hovy, CMU
2023	**PhD** *Dylan Ebert*	Ellie Pavlick, Brown
2023	**MSR** *Mateo Guaman Castro*	Sebastian Scherer, CMU
2022	**PhD** *Shikib Mehri*	Maxine Eskenazi, CMU
2022	**PhD** *Shruti Palaskar*	Alan Black and Florian Metze, CMU
2022	**PhD** *Evangelia Spiliopoulou*	Ed Hovy, CMU
2022	**PhD** *Dheeraj Rajagopal*	Ed Hovy, CMU
2022	**PhD** *Volkan Cirik*	Louis-Philippe Morency and Taylor Berg-Kirkpatrick, CMU
2022	**MLT** *Aishwarya Jadhav*	Katia Sycara, CMU
2021	**PhD** *Pengcheng Yin*	Graham Neubig, CMU
2021	**MLT** *Jing Wen*	Katerina Fragkiadaki, CMU

2020 **PhD** *Zihang Dai* Yiming Yang, CMU

Teaching

2027 **11-851 Talking to Robots**, SP
2026 **11-977 Online MultiModal Machine Learning Certificate**, SP
2025 **11-851 Talking to Robots** # Students: 38 , FA
– **11-777 MultiModal Machine Learning** # Students: 145, SP
2024 **11-851 Talking to Robots** # Students: 37 , FA
– **11-777 MultiModal Machine Learning** *w/ Daniel Fried* # Students: 83 , FA
– **11-777 MultiModal Machine Learning** # Students: 135, SP
2023 **11-851 Talking to Robots (created)** # Students: 16 , FA
– **11-767 On-Device Machine Learning** *w/ Emma Strubell* # Students: 69 , FA
– **11-777 MultiModal Machine Learning** *w/ Daniel Fried* # Students: 107, SP
2022 **11-767 On-Device Machine Learning** *w/ Emma Strubell* # Students: 42 , FA
– **11-777 MultiModal Machine Learning** # Students: 94 , SP
2021 **11-767 On-Device Machine Learning (co-created)** *w/ Emma Strubell* # Students: 31 , FA
– **11-777 MultiModal Machine Learning** # Students: 42 , SP

(Senior) Area Chair / Action Editor

2027 **AC: IEEE International Conference on Robotics and Automation (ICRA)** Robo
– **AC: International Conference on Machine Learning (ICML)** AI/ML
2026 **AC: Conference on Language Modeling (COLM)** NLP
– **AC: Conference on Robot Learning (CoRL)** Robo
– **AE: Transactions on Machine Learning Research (TMLR)** AI/ML
2025 **AC: IEEE International Conference on Robotics and Automation (ICRA)** Robo
– **AC: Conference on Robot Learning (CoRL)** Robo
– **AC: Robotics: Science and Systems (RSS)** Robo
– **AC: Conference on Language Modeling (COLM)** NLP
– **SAE: ACL Rolling Review (ARR/EMNLP)** NLP
– **AC: International Conference on Machine Learning (ICML)** AI/ML
– **AE: Transactions on Machine Learning Research (TMLR)** AI/ML
2024 **AC: Conference on Robot Learning (CoRL)** Robo
– **AC: Conference on Computer Vision and Pattern Recognition (CVPR)** CV
– **AC: International Conference on Learning Representations (ICLR)** AI/ML
– **AE: Transactions on Machine Learning Research (TMLR)** AI/ML
– **AC: European Chapter of the Association for Computational Linguistics (EACL)** NLP
– **SAE: ACL Rolling Review (ARR)** NLP
2023 **AC: Neural Information Processing Systems (NeurIPS)** AI/ML
– **AC: International Conference on Machine Learning (ICML)** AI/ML
– **AE: Transactions on Machine Learning Research (TMLR)** AI/ML
– **SAE: ACL Rolling Review (ARR)** NLP
– **Ethics Co-Chair: Association for Computational Linguistics (ACL)** NLP
– **SAC: Empirical Methods in Natural Language Proc (EMNLP)** NLP
2022 **AC: International Conference on Learning Representations (ICLR)** AI/ML
– **AC: Neural Information Processing Systems (NeurIPS)** AI/ML
– **Senior MR: Association for the Advancement of Artificial Intelligence (AAAI)** AI/ML
– **AE: Transactions on Machine Learning Research (TMLR)** AI/ML
– **SAC: Empirical Methods in Natural Language Proc (EMNLP)** NLP

–	**Senior AE: ACL Rolling Review (ARR)**	NLP
2021	**AC: International Conference on Learning Representations (ICLR)**	AI/ML
–	**AC: Neural Information Processing Systems (NeurIPS)**	AI/ML
–	**AC: Association for Computational Linguistics (ACL)**	NLP
–	**AC: Empirical Methods in Natural Language Proc (EMNLP)**	NLP
–	**AE: ACL Rolling Review (ARR)**	NLP
2020	**AC: Association for Computational Linguistics (ACL)**	NLP
–	**AC: Asia-Pacific chapter of the Assoc. for Computational Linguistics (AACL)**	NLP
–	**AC: Neural Information Processing Systems (NeurIPS)**	AI/ML
2019	**AC: Association for Computational Linguistics (ACL)**	NLP
2018	**AC: International Conference on Computational Linguistics**	NLP
2016	**AC: Conference on Lexical and Computational Semantics (*SEM)**	NLP

Reviewer

NLP	**ARR**	Association for Computational Linguistics Rolling Review	2021
	ACL	Association for Computational Linguistics	2017, 2018
	CL	Computational Linguistics Journal	2017 – 2022
	CoLing	International Conference on Computational Linguistics	2016
	CoNLL	Conference on Computational Natural Language Learning	2016 – 2018
	EACL	European Chapter of the Association for Computational Linguistics	2014, 17, 20
	EMNLP	Empirical Methods in Natural Language Processing	2013 – 2020
	–	Best Papers Awards Committee	2024
	IJCNLP	International Joint Conference on Natural Language Processing	2017
	NAACL	N. American Chapter of the Association for Computational Ling	2016 – 2019
	–	Best Papers Awards Committee	2025
	NLE	Natural Language Engineering Journal	2016
	***SEM**	Joint Conference on Lexical and Computational Semantics	2020
	TACL	Transactions of the Assoc. for Computational Linguistics	2019 – 2023
Robo	**CoRL**	Conference on Robot Learning	2019 – 2021,2023
	AURO	Autonomous Robots	2022
	RSS	Robotics: Science and Systems	2022, 2024
	HRI	Human Robot Interaction	2020 – 2025
	ICRA	International Conference on Robotics and Automation	2019,2023 – 25
	IJRR	International Journal of Robotics Research	2017
	IROS	IEEE/RSJ International Conference on Intelligent Robots and Systems	2019,23–24
	RA-L	IEEE Robotics and Automation Letters	2019,2023,2026
AI/ML	**AAAI**	Association for the Advancement of Artificial Intelligence	2018, 2021
	ICLR	International Conference on Learning Representations	'19 – '20,'25,'27
	ICML	International Conference on Machine Learning	2019
	IJCAI	International Joint Conference on Artificial Intelligence	2019
	NeurIPS	Neural Information Processing Systems	2018 – 2019,2025
	T-PAMI	Transactions on Pattern Analysis and Machine Intelligence	2023 –
	TMLR	Transactions on Machine Learning Research	2026 –
CV	**CVPR**	Computer Vision and Pattern Recognition	2020, 2023
	ICCV	International Conference on Computer Vision	2025
	WACV	IEEE/CVF Winter Conference on Applications of Computer Vision	2026

Organization (Workshops and Conference)

2022 **Workshop** *Embodied AI Workshop – Challenge Organizer*
Computer Vision and Pattern Recognition

2021 **Workshop** *Embodied AI Workshop – Challenge Organizer*
Computer Vision and Pattern Recognition

2020 **Workshop** *Embodied Vision, Actions & Language Workshop*
European Conference on Computer Vision

– **Workshop** *Workshop on Dataset Curation and Security*
NeurIPS

2019 **Workshop** *Spatial Language Understanding and Grounded communication for Robotics*
North American Chapter of the Association for Computational Linguistics

2018 **Conference** *Social Media Chair*
North American Chapter of the Association for Computational Linguistics

– **Workshop** *Generalization in the Age of Deep Learning*
North American Chapter of the Association for Computational Linguistics

– **Workshop** *Wordplay: Reinforcement and Language Learning in Text-Based Games*
NeurIPS

2017 **Conference** *Student Scholarship Chair and Student Volunteer Coordinator*
Conference on Empirical Methods in Natural Language Processing

– **Workshop** *Language Grounding for Robotics*
Conference of the Association for Computational Linguistics

Invitations and Talks

2026 **Language is Discrete, the World is Continuous** *Japanese-American-German Frontiers of Science*, Japan Society for the Promotion of Science, the US National Academy of Sciences and the Alexander von Humboldt Foundation

Welcome to Symbols *IROS Workshop on AI Meets Autonomy*

2025 **Embodied Language: Evaluating LLMs in the Real World** *75th Anniversary of the Turing Test*, Université du Québec à Montréal

Spatial and Semantic Awareness in Embodied LMs *IVADO 1st Workshop: Assessing and Improving the Capabilities and Safety of Agents*, Montreal

Guiding and Calibrating LLMs *Apple Workshop on Human-Centered Machine Learning*

Semantic Spaces and VLM limitations *Stanford Vision and Learning Lab (SVL)*

Following Instructions and Asking Questions *Army Research Lab*

Why is language an embodied problem? *UK Robotics Summer School Edinburgh*

Planning, Reasoning, and Generalization: A title too bold for me to actually deliver on *JP Morgan AI Agents Seminar*

Semantics? Reasoning? Can we define either of those terms? *RSS Workshop: 2nd Workshop on Semantic Reasoning and Goal Understanding in Robotics*

Humans Know Things, Humans Get in the Way *RSS Workshop: Continual Robot Learning from Humans*

Simulating People with Language Models – Good Idea? Bad Idea? *RSS Workshop: GenAI-HRI*

AI: Powerful and Limited *Next Gen Philanthropy*

2024 **Talking to Robots** *University Pennsylvania*

Everything Fails, Everything is Ambiguous *RSS Social Intelligence Workshop*

Do It Like This, and Don't Wake the Baby *RSS Task Specification Workshop*

University of Maryland, College Park

2023 **Are We Really Talking to Robots?** *"Dialogue with Robots" workshop*, UPitt NSF Workshop

Words ≠ language *Articulate Robots: Utilizing Language for Robot Learning*, RSS Workshop

Following Instructions and Asking Questions *Learning by Asking for Intelligent Robots and Agents*, RO-MAN Workshop

Following Instructions and Asking Questions *Goal Conditioned RL Workshop @ NeurIPS*

2022 **Can we *really* connect language to robotics?** *CoRL Wksp on Language and Robot Learning*

How machines learn language: Useful but wrong? *12th Inter. Wksp on Language Production*

Following Instructions and Asking Questions *Univ of Potsdam*

Univ California Santa Barbara NLP
Georgia Tech NLP
Microsoft Research NYC
Penn State University
Universitat Hamburg, Crossmodal Learning (CML) summer school
University of Stuttgart, Institute of Natural Language Processing
University of Pittsburgh NLP

2021 *Univ of Illinois at Urbana-Champaign NLP*
Human Interaction for Robotic Navigation Workshop @ Inter. Conference on Computer Vision
WebQA: Multihop and Multimodal QA *PNNL Mega-AI Seminar*
WebQA: Multihop and Multimodal *Microsoft Research Summit*
2nd Workshop on Advances in Language and Vision Research (ALVR) *NAACL*
Instructions, Abstraction, and Theory-of-Mind *Koç University AI center*
Understanding the Potential of Scaling Artificial Intelligence for Science Applications Panel *Pacific Northwest National Lab*
Guest Lecture: Embodiment for NLP *Univ of Pennsylvania*
Guest Lecture: Multimodality in NLP *UWashington Linguistics*

2020 **The 2nd Workshop Beyond Vision and Language: Integrating Real-world Knowledge** *CoLing*
The University of Texas at Austin – Forum for AI
Cornell Learning Machines seminar
Stanford NLP seminar
SRI Center for Vision Technology Lab Panel *Embodied AI*

2018 **Colloquium Talk** *Northwest NLP*

2017 **Invited Talk** *University of Notre Dame NLP Seminar*
Invited Talk *University of California, Irvine AI/ML Seminar*
Invited Talk *Pacific Northwest National Laboratory*

2016 **Invited Poster** *The Impact of Computing in Our Physical World session of the CRA's Computing Community Consortium (CCC) Symposium on Computing Research*

2015 **Colloquium Talk** *Toyota Tech. Inst. of Chicago's Midwest Speech & Language Days*

2013 **Colloquium Talk** *Toyota Tech. Inst. of Chicago's Midwest Speech & Language Days*